AI ASSISTANT CODING
ASSIGNMENT-17.2

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Batch: 34

Semester: VI

Branch: CSE(AIML)

Lab 17 – AI for Data Processing: Data Cleaning and Preprocessing Scripts

Task Description – 1(Student Academic Data Cleaning)

• Task: Use AI assistance to generate a Python script that cleans and prepares a student academic dataset for analysis.

Instructions:

• Handle missing values in marks, attendance, and department.

• Remove duplicate student records based on student ID.

• Standardize text fields such as department names.

• Encode categorical variables like gender and department.

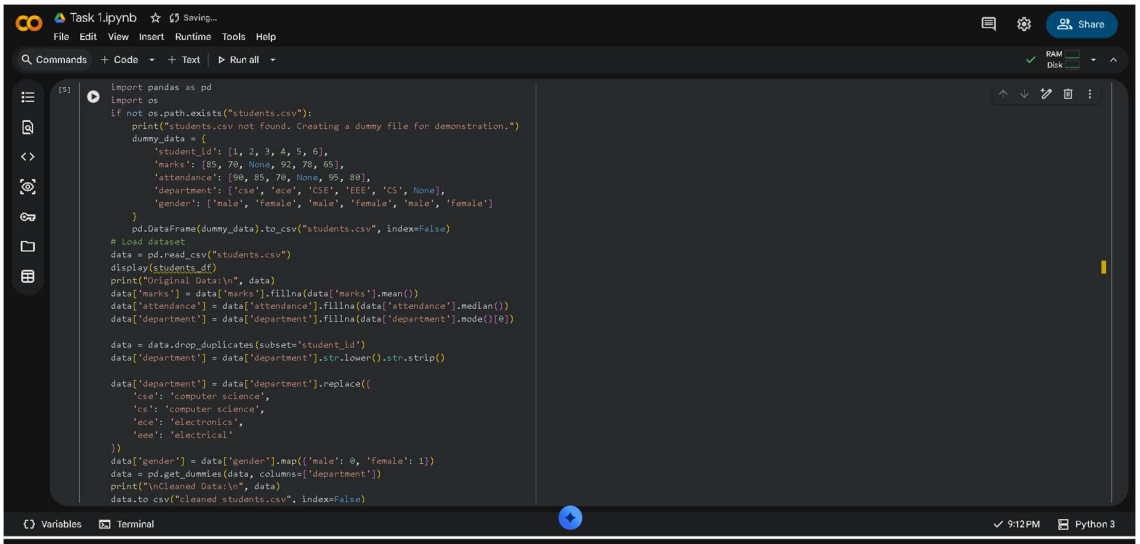
Sample Code:

```
import pandas as pd

data = pd.read_csv("students.csv")

print(data.head())
```

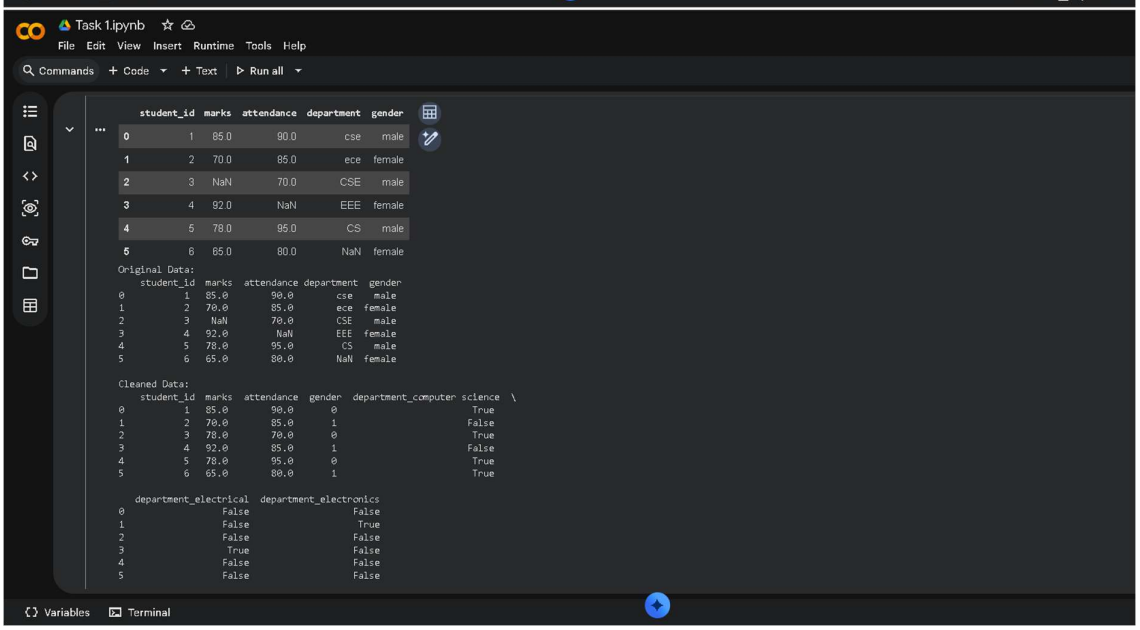
Expected Output:

• A cleaned and encoded Pandas DataFrame suitable for academic

performance analysis.



```python
import pandas as pd
import os
if not os.path.exists("students.csv"):
    print("students.csv not found. Creating a dummy file for demonstration.")
    dummy_data = {
        'student_id': [1, 2, 3, 4, 5, 6],
        'marks': [85, 70, None, 92, 78, 65],
        'attendance': [90, 85, 70, None, 95, 80],
        'department': ['cse', 'ece', 'CSE', 'EEE', 'CS', None],
        'gender': ['male', 'female', 'male', 'female', 'male', 'female']
    }
    pd.DataFrame(dummy_data).to_csv("students.csv", index=False)
# Load dataset
data = pd.read_csv("students.csv")
display(students_df)
print("Original Data:\n", data)
data['marks'] = data['marks'].fillna(data['marks'].mean())
data['attendance'] = data['attendance'].fillna(data['attendance'].median())
data['department'] = data['department'].fillna(data['department'].mode()[0])

data = data.drop_duplicates(subset='student_id')
data['department'] = data['department'].str.lower().str.strip()

data['department'] = data['department'].replace({
    'cse': 'computer science',
    'cs': 'computer science',
    'ece': 'electronics',
    'eee': 'electrical'
})
data['gender'] = data['gender'].map({'male': 0, 'female': 1})
data = pd.get_dummies(data, columns=['department'])
print("\nCleaned Data:\n", data)
data.to_csv("cleaned_students.csv", index=False)
```



	student_id	marks	attendance	department	gender
0	1	85.0	90.0	cse	male
1	2	70.0	85.0	ece	female
2	3	NaN	70.0	CSE	male
3	4	92.0	NaN	EEE	female
4	5	78.0	95.0	CS	male
5	6	65.0	80.0	NaN	female

```
Original Data:
   student_id  marks  attendance department  gender
0           1   85.0        90.0        cse    male
1           2   70.0        85.0        ece  female
2           3    NaN        70.0        CSE    male
3           4   92.0         NaN        EEE  female
4           5   78.0        95.0         CS    male
5           6   65.0        80.0        NaN  female

Cleaned Data:
   student_id  marks  attendance  gender  department_computer science  \
0           1   85.0        90.0       0                         True
1           2   70.0        85.0       1                        False
2           3   78.0        70.0       0                         True
3           4   92.0        85.0       1                        False
4           5   78.0        95.0       0                         True
5           6   65.0        80.0       1                         True

   department_electrical  department_electronics
0                  False                   False
1                  False                    True
2                  False                   False
3                   True                   False
4                  False                   False
5                  False                   False
```

Task Description – 2 (Financial Transaction Data Preprocessing)

- **Task:** Preprocess financial transaction data using AI-generated Python code.

Instructions:

- Convert transaction date columns to datetime format.
- Create derived features such as transaction month and year.
- Normalize the transaction amount column.
- Identify and handle extreme transaction values.

Sample Code:

import pandas as pd

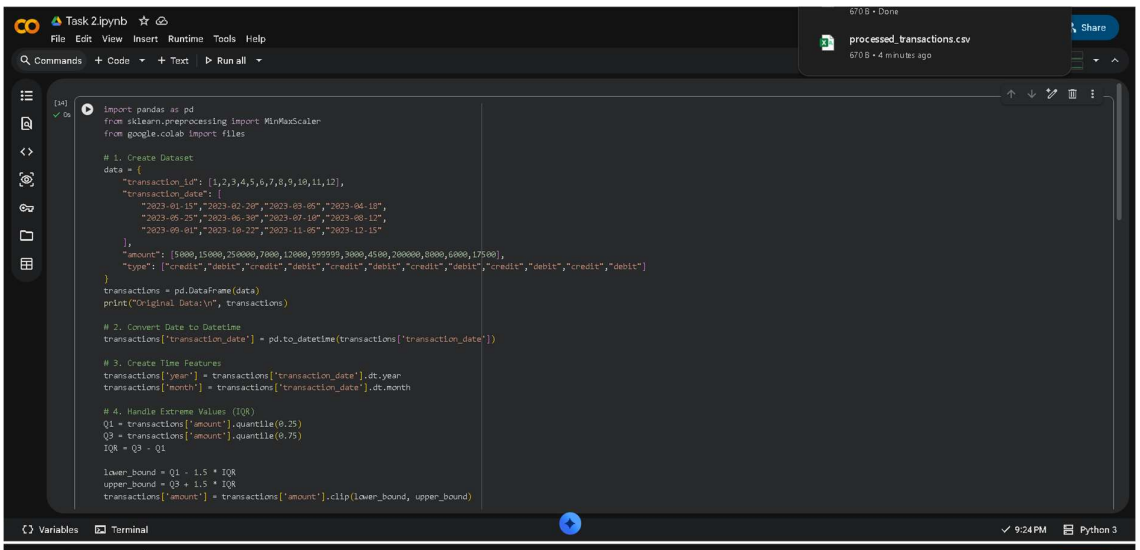
from sklearn.preprocessing import MinMaxScaler

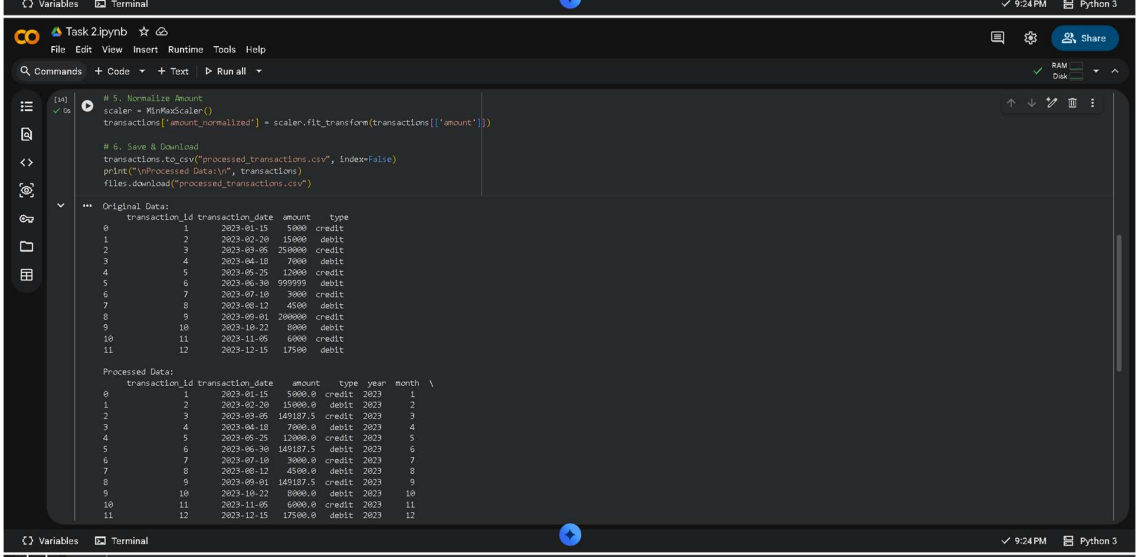
transactions = pd.read_csv("transactions.csv")

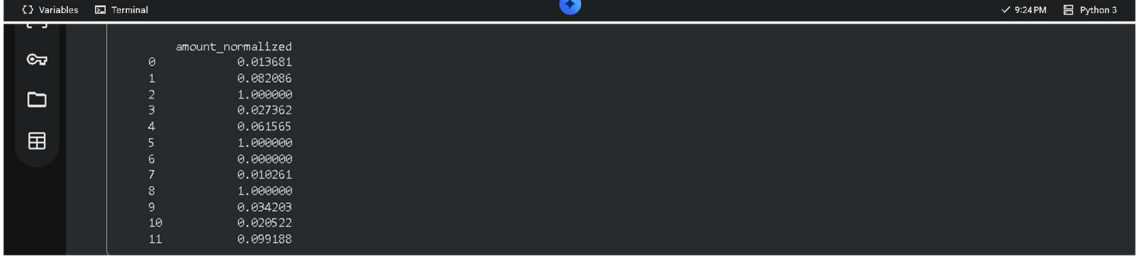
print(transactions.info())

Expected Output:

• A transformed dataset with normalized values and newly generated time based features.







	amount_normalized
0	0.013681
1	0.082086
2	1.000000
3	0.027362
4	0.061565
5	1.000000
6	0.000000
7	0.010261
8	1.000000
9	0.034203
10	0.020522
11	0.099188

Task 3 – (Environmental Sensor Data Preparation)

• Task: Clean and preprocess environmental sensor data using AI-assisted scripts.

Instructions:

• Handle missing values in temperature, humidity, and air_quality_index.

• Scale numerical features using standard scaling.

• Encode categorical fields such as sensor_status.

• Split the dataset into training and testing subsets.

Sample Code:

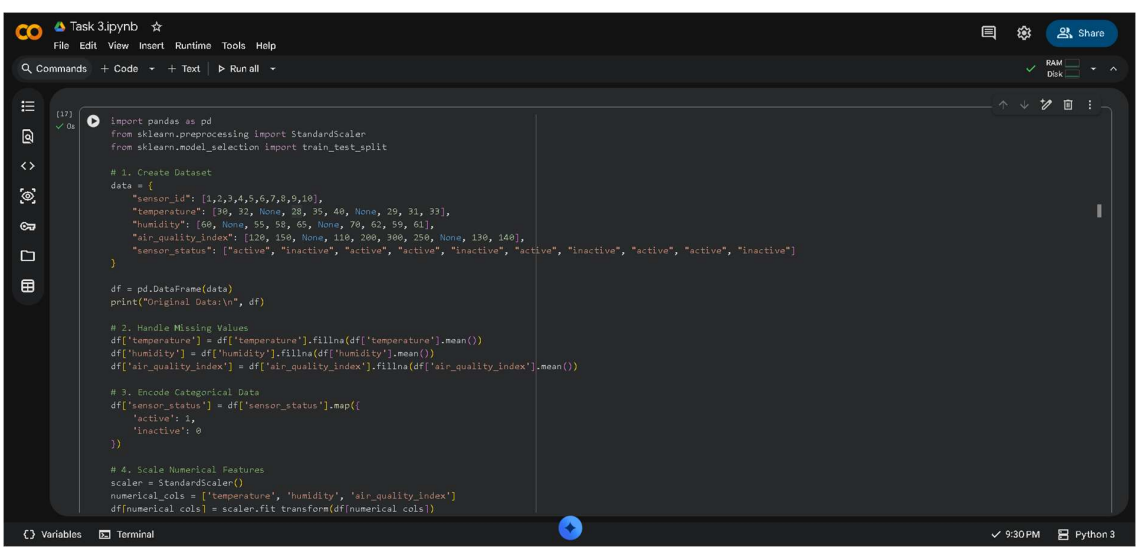
import pandas as pd

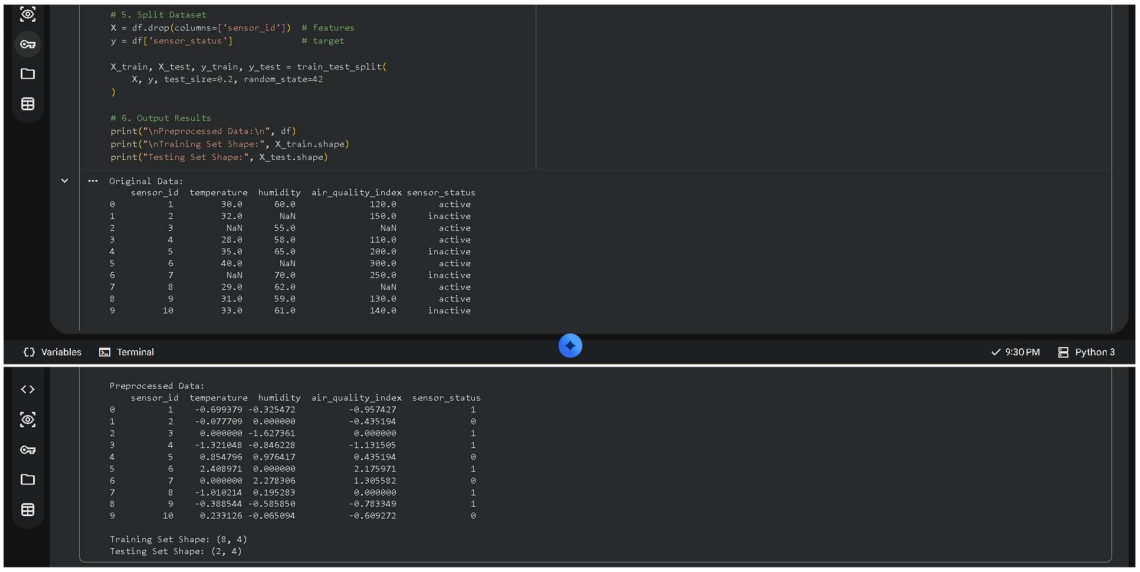
from sklearn.model_selection import train_test_split

data = pd.read_csv("sensor_data.csv")

Expected Output:

• A fully preprocessed dataset ready for predictive modeling tasks.





```python
# 5. Split Dataset
X = df.drop(columns=['sensor_id'])  # features
y = df['sensor_status']             # target

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

# 6. Output Results
print("\nPreprocessed Data:\n", df)
print("\nTraining Set Shape:", X_train.shape)
print("Testing Set Shape:", X_test.shape)
```

```
Original Data:
   sensor_id  temperature  humidity  air_quality_index sensor_status
0          1         30.0      60.0              120.0        active
1          2         32.0       NaN              150.0      inactive
2          3          NaN      55.0                NaN        active
3          4         28.0      58.0              110.0        active
4          5         35.0      65.0              200.0      inactive
5          6         40.0       NaN              300.0        active
6          7          NaN      70.0              250.0      inactive
7          8         29.0      62.0                NaN        active
8          9         31.0      59.0              130.0        active
9         10         33.0      61.0              140.0      inactive
```

Variables Terminal 9:30 PM Python 3

```
Preprocessed Data:
   sensor_id  temperature  humidity  air_quality_index  sensor_status
0          1    -0.699379 -0.325472          -0.957427              1
1          2    -0.077709  0.000000          -0.435194              0
2          3     0.000000 -1.627361           0.000000              1
3          4    -1.321048 -0.846228          -1.131505              1
4          5     0.854796  0.976417           0.435194              0
5          6     2.408971  0.000000           2.175971              1
6          7     0.000000  2.278306           1.305582              0
7          8    -1.010214  0.195283           0.000000              1
8          9    -0.388544 -0.585850          -0.783349              1
9         10     0.233126 -0.065094          -0.609272              0

Training Set Shape: (8, 4)
Testing Set Shape: (2, 4)
```

Task 4 – (Real-Time Application: Smart City Traffic Data Cleaning)

• Scenario:

A smart city system collects traffic data from multiple sensors, resulting in noisy and inconsistent records.

Requirements:

• Standardize road and location names.

• Fill missing traffic density values using appropriate statistical methods.

• Remove duplicate sensor readings.

• Normalize speed and vehicle count metrics.

• Generate a brief summary comparing dataset quality before and after cleaning

Sample Code:

```
import pandas as pd

traffic = pd.read_csv("traffic_data.csv")

print(traffic.describe())
```

Expected Output:

• A cleaned and normalized traffic dataset with an accompanying

data-quality improvement summary.

Task 4.ipynb

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[18]

```python
import pandas as pd
from sklearn.preprocessing import MinMaxScaler

# 1. Create Sample Dataset
data = {
    "sensor_id": [1,2,3,4,5,6,7,8,9,10,2],
    "road_name": ["MG Road", "mg road ", "M.G. Road", "Highway 1", "highway-1",
                  "Main St", "main street", "MAIN ST", "Ring Rd", "ring road", "mg road"],
    "location": ["Downtown", "downtown ", "DOWN TOWN", "Uptown", "uptown",
                 "Midtown", "mid town", "MIDTOWN", "Suburb", "suburb", "downtown"],
    "traffic_density": [50, None, 55, 80, None, 40, 42, None, 60, 65, 50],
    "speed": [45, 50, 48, 60, 62, 35, 38, 36, 55, 58, 50],
    "vehicle_count": [200, 220, 210, 400, 420, 150, 160, 155, 300, 320, 220]
}

traffic = pd.DataFrame(data)
print("🔴 Original Data:\n", traffic)

# 2. Data Quality BEFORE Cleaning
print("\n🔍 BEFORE CLEANING SUMMARY")
print("Missing Values:\n", traffic.isnull().sum())
print("Duplicate Rows:", traffic.duplicated().sum())

# 3. Standardize Text Fields
traffic['road_name'] = traffic['road_name'].str.lower().str.strip()
traffic['location'] = traffic['location'].str.lower().str.strip()

# Normalize variations
traffic['road_name'] = traffic['road_name'].replace({
    "mg road": "mg road",
    "m.g. road": "mg road",
```

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Task 4.ipynb

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[18]

```python
# Normalize variations
traffic['road_name'] = traffic['road_name'].replace({
    "mg road": "mg road",
    "m.g. road": "mg road",
    "mg road ": "mg road",
    "highway-1": "highway 1",
    "main street": "main st",
    "main st": "main st",
    "ring road": "ring rd"
})
traffic['location'] = traffic['location'].replace({
    "down town": "downtown",
    "downtown ": "downtown",
    "mid town": "midtown"
})

# 4. Handle Missing Values
traffic['traffic_density'] = traffic['traffic_density'].fillna(
    traffic['traffic_density'].median()
)

# 5. Remove Duplicates
traffic = traffic.drop_duplicates()

# 6. Normalize Numerical Features
scaler = MinMaxScaler()
traffic[['speed', 'vehicle_count']] = scaler.fit_transform(
    traffic[['speed', 'vehicle_count']]
)

# 7. Data Quality AFTER Cleaning
print("\n🟢 AFTER CLEANING SUMMARY")
print("Missing Values:\n", traffic.isnull().sum())
```

Variables Terminal 9:34 PM Python 3

Task 4.ipynb

File Edit View Insert Runtime Tools Help

Commands + Code + Text Run all

[18]

```python
# 7. Data Quality AFTER Cleaning
print("\n🟢 AFTER CLEANING SUMMARY")
print("Missing Values:\n", traffic.isnull().sum())
print("Duplicate Rows:", traffic.duplicated().sum())

# 8. Final Output
print("\n✅ Cleaned Data:\n", traffic)

# Save cleaned dataset
traffic.to_csv("cleaned_traffic_data.csv", index=False)
```

```
🔴 Original Data:
   sensor_id    road_name   location  traffic_density  speed  vehicle_count
0          1      MG Road   Downtown             50.0     45            200
1          2     mg road   downtown              NaN     50            220
2          3    M.G. Road  DOWN TOWN             55.0     48            210
3          4    Highway 1     Uptown             80.0     60            400
4          5    highway-1     uptown              NaN     62            420
5          6      Main St    Midtown             40.0     35            150
6          7  main street   mid town             42.0     38            160
7          8      MAIN ST   MIDTOWN              NaN     36            155
8          9      Ring Rd     Suburb             60.0     55            300
9         10    ring road     suburb             65.0     58            320
10         2     mg road   downtown             50.0     50            220

🔍 BEFORE CLEANING SUMMARY
Missing Values:
 sensor_id        0
road_name        0
location         0
traffic_density  3
speed            0
vehicle_count    0
dtype: int64
Duplicate Rows: 0
```

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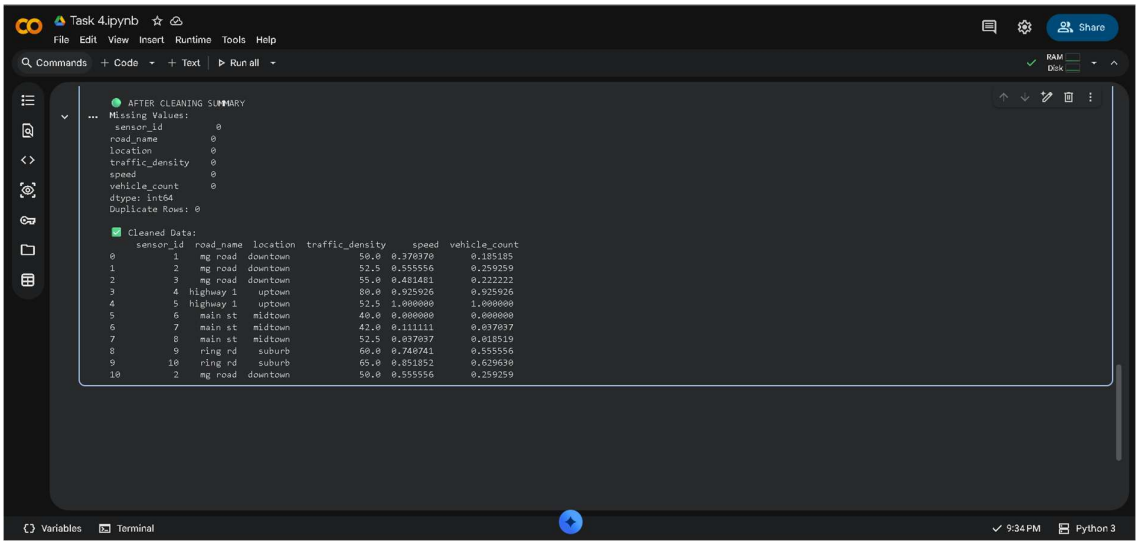

Task 5 – (Social Media Text Data Cleaning and Feature Extraction)

• Task:

Use AI-assisted Python scripting to clean and preprocess social media text data for sentiment analysis.

Instructions:

• Remove duplicate posts and null text entries.

• Clean text by removing URLs, special characters, and extra spaces.

• Convert text to lowercase and perform basic tokenization.

• Remove stopwords and perform stemming or lemmatization. Generate basic text features such as word count and sentiment score.

Sample Code:

```
import pandas as pd

import re

posts = pd.read_csv("social_media_posts.csv")

print(posts.head())
```

Expected Output:

A cleaned text dataset with extracted features suitable for sentiment analysis models.

Task 5.ipynb

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[19]

```
ipip install nltk textblob
```

```
Requirement already satisfied: nltk in /usr/local/lib/python3.12/dist-packages (3.9.1)
Requirement already satisfied: textblob in /usr/local/lib/python3.12/dist-packages (0.19.0)
Requirement already satisfied: click in /usr/local/lib/python3.12/dist-packages (from nltk) (8.3.1)
Requirement already satisfied: joblib in /usr/local/lib/python3.12/dist-packages (from nltk) (1.5.3)
Requirement already satisfied: regex>=2021.8.3 in /usr/local/lib/python3.12/dist-packages (from nltk) (2025.11.3)
Requirement already satisfied: tqdm in /usr/local/lib/python3.12/dist-packages (from nltk) (4.67.3)
```

[21]

```
import pandas as pd
import re
from sklearn.feature_extraction.text import ENGLISH_STOP_WORDS
from nltk.stem import PorterStemmer
from textblob import TextBlob

# 1. Create Sample Dataset
data = {
    "post_id": [1,2,3,4,5,6,7,8,2],
    "text": [
        "I love this product! 😊 http://example.com",
        "Worst service ever!!! 😡",
        None,
        "Amazing experience, will buy again!!!",
        "Check this out: https://test.com #awesome",
        "Not good, very bad service",
        "OK product, nothing special",
        "I LOVE THIS PRODUCT! 😊 http://example.com",
        "Worst service ever!!! 😡"
    ]
}
```

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Task 5.ipynb

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Commands + Code + Text Run all

[21]

```
posts = pd.DataFrame(data)
print("🔴 Original Data:\n", posts)

# 2. Remove Duplicates & Nulls
posts = posts.drop_duplicates(subset='text')
posts = posts.dropna(subset=['text'])

# 3. Clean Text
def clean_text(text):
    text = text.lower()
    text = re.sub(r"http\S+", "", text)        # remove URLs
    text = re.sub(r"[^a-z\s]", "", text)       # remove special chars
    text = re.sub(r"\s+", " ", text).strip()   # remove extra spaces
    return text

posts['clean_text'] = posts['text'].apply(clean_text)

# 4. Tokenization
posts['tokens'] = posts['clean_text'].apply(lambda x: x.split())

# 5. Remove Stopwords
posts['tokens'] = posts['tokens'].apply(
    lambda words: [w for w in words if w not in ENGLISH_STOP_WORDS]
)

# 6. Stemming
stemmer = PorterStemmer()
posts['tokens'] = posts['tokens'].apply(
    lambda words: [stemmer.stem(w) for w in words]
)

# Join tokens back
posts['processed_text'] = posts['tokens'].apply(lambda x: " ".join(x))
```

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Task 5.ipynb

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Commands + Code + Text Run all

[21]

```
# Join tokens back
posts['processed_text'] = posts['tokens'].apply(lambda x: " ".join(x))

# 7. Feature Extraction

# Word count
posts['word_count'] = posts['processed_text'].apply(lambda x: len(x.split()))

# Sentiment score
posts['sentiment_score'] = posts['processed_text'].apply(
    lambda x: TextBlob(x).sentiment.polarity
)

# 8. Final Output
print("\n🟢 Cleaned & Processed Data:\n", posts)

# Save file
posts.to_csv("cleaned_social_media_posts.csv", index=False)
```

```
🔴 Original Data:
   post_id                                       text
0        1  I love this product! 😊 http://example.com
1        2                     Worst service ever!!! 😡
2        3                                        None
3        4      Amazing experience, will buy again!!!
4        5  Check this out: https://test.com #awesome
5        6                   Not good, very bad service
6        7                  OK product, nothing special
7        8  I LOVE THIS PRODUCT! 😊 http://example.com
8        2                     Worst service ever!!! 😡

🟢 Cleaned & Processed Data:
   post_id                                       text  \
0        1  I love this product! 😊 http://example.com
```

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Task 5.ipynb

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Commands + Code + Text Run all

```
🔴 Original Data:
   post_id                                       text
0        1  I love this product! 😊 http://example.com
1        2                        Worst service ever!!! 😡
2        3                                           None
3        4       Amazing experience, will buy again!!!
4        5  Check this out: https://test.com #awesome
5        6                  Not good, very bad service
6        7                 OK product, nothing special
7        8  I LOVE THIS PRODUCT! 😊 http://example.com
8        2                        Worst service ever!!! 😡

🟢 Cleaned & Processed Data:
   post_id                                       text  \
0        1  I love this product! 😊 http://example.com
1        2                        Worst service ever!!! 😡
3        4       Amazing experience, will buy again!!!
4        5  Check this out: https://test.com #awesome
5        6                  Not good, very bad service
6        7                 OK product, nothing special
7        8  I LOVE THIS PRODUCT! 😊 http://example.com

                          clean_text                   tokens  \
0                i love this product         [love, product]
1                  worst service ever         [worst, servic]
3  amazing experience will buy again      [amaz, experi, buy]
4             check this out awesome          [check, awesom]
5            not good very bad service     [good, bad, servic]
6          ok product nothing special  [ok, product, special]
7                i love this product         [love, product]

       processed_text  word_count  sentiment_score
0       love product           2     5.000000e-01
1       worst servic           2    -1.000000e+00
3    amaz experi buy           3     0.000000e+00
4      check awesom            2     0.000000e+00
```

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```
       processed_text  word_count  sentiment_score
0       love product           2     5.000000e-01
1       worst servic           2    -1.000000e+00
3    amaz experi buy           3     0.000000e+00
4      check awesom            2     0.000000e+00
5    good bad servic           3     5.551115e-17
6  ok product special          3     4.285714e-01
7       love product           2     5.000000e-01
```